Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/04/2021
PI Name:	Alfano, Candice Ph.D.		
Project Title:	Characterization of Psychological Risk, Overlap with Physical Health, and Associated Performance in Isolated, Confined, Extreme (ICE) Environments		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	caalfano@uh.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-8611
Organization Name:	University of Houston		
PI Address 1:	Psychology Department		
PI Address 2:	126 Heyne Bldg		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-5022	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-BMED Behavioral Health & Performance
Start Date:	11/13/2014	**End Date:**	09/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2022 per NSSC information (Ed., 8/24/21) NOTE: End date changed to 9/30/2021 per NSSC information (Ed., 4/22/2020) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 1/29/2020) NOTE: End date changed to 12/31/2019 per NSSC information (Ed., 7/17/19) NOTE: End date changed to 5/31/2019 per NSSC information (Ed., 2/12/19) NOTE: End date changed to 1/31/2019 per NSSC information (Ed., 12/28/18) NOTE: End date changed to 11/12/2018 per NSSC information (Ed., 12/13/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17)		
Key Personnel Changes/Previous PI:	July 2021 report: Dr. Matthew Gallagher was added as a Co-Investigator-statistician during the NCE starting in April 2020; Dr. Mitzi Laughlin is no longer CoI on the project.		

COI Name (Institution):	Connaboy, Christopher Ph.D. (University of Pittsburgh) Simpson, Richard Ph.D. (University of Houston) Gallagher, Matthew Ph.D. (University of Houston)
Grant/Contract No.:	NNX15AC13G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Anecdotal and empirical findings collected in space and other extreme environments continue to highlight the potential for psychological symptoms and conditions to degrade crew performance, increase conflict, and jeopardize mission success. Indeed, 'negative reactions' during periods of isolation, confinement, demanding work schedules, stimulus reduction, separation from loved ones, sleep deprivation, and a host of other stressors are more appropriately viewed as normative rather than pathogenic. Selection methods and countermeasures serve to mitigate some degree of psychological risk, but long-duration space flight will substantially extend exposure to these and other stressors. Previous research documenting psychological symptoms experienced during space flight and in other isolated and confined environments (ICE) provides evidence of a wide range of psychological and behavioral reactions. Unfortunately however, these collective data ultimately serve to raise more questions than answers. Differences in collection methods, types of symptoms/reactions assessed, psychological constructs examined, and timing and duration of measurements limit conclusions that can be drawn from this research. As a result, understanding of the discrete symptoms and conditions most likely to occur during space flight and thus, ability to quantify the magnitude, probability, or consequences of such risk remains inadequate. The current project proposes to address these notable gaps in knowledge via three specific Aims. First, we will conduct extensive scientific literature reviews and interviews with subject matter experts in order to synthesize existing knowledge of the psychological and behavioral symptoms experienced in space and other extreme environments (Aim 1). Our review will directly inform the development of a comprehensive checklist of symptoms to be monitored among 6 separate cohorts (i.e., 2 Antarctic and 4 Human Exploration Research Analog (HERA) cohorts)) as part of a longitudinal investigation (Aim 2). Symptoms will be examined based on their point/period prevalence, severity, and duration. The checklist will also be administered (repeatedly) in conjunction with a physical symptoms checklist in order to examine concurrent and sequential overlap between psychological and physical health symptoms as means of clarifying potential etiologies. Finally, our study will extend previous research by exploring relationships among psychological health, sleep loss/dysregulation, biomarkers of stress, and performance-based outcomes (Aim 3). A comprehensive battery of cognitive and performance measures (including a perception-action coupling task) will be administered repeatedly as part of our longitudinal study. These outcomes will inform a final list of psychological/ behavioral symptoms to be examined during an extended International Space Station (ISS) mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Anecdotal and empirical findings collected in space and other extreme environments continue to highlight the potential for psychological symptoms and conditions to degrade crew performance, increase conflict, and jeopardize mission success. Indeed, 'negative reactions' during periods of isolation, confinement, demanding work schedules, stimulus reduction, separation from loved ones, sleep deprivation, and a host of other stressors are more appropriately viewed as normative rather than pathogenic. Selection methods and countermeasures serve to mitigate some degree of psychological risk, but long-duration space flight will substantially extend exposure to these and other stressors. Previous research documenting psychological symptoms experienced during space flight and in other isolated and confined environments (ICE) provides evidence of a wide range of psychological and behavioral reactions. Unfortunately however, these collective data ultimately serve to raise more questions than answers. Differences in collection methods, types of symptoms/reactions assessed, psychological constructs examined, and timing and duration of measurements limit conclusions that can be drawn from this research. As a result, understanding of the discrete symptoms and conditions most likely to occur during space flight and thus, ability to quantify the magnitude, probability, or consequences of such risk remains inadequate. Thus, there is a need to: (1) identify the psychological/behavioral symptoms that pose the greatest threat to performance; (2) provide accurate and acceptable risk thresholds; (3) inform screening and selection processes; (4) guide further development of suitable working practices (standard operating procedures); and (5) develop interventions and counter measures to mitigate these risks. This project specifically addresses several knowledge gaps related to Risks of Adverse Behavioral Conditions and Psychiatric Disorders [Ed. note: August 2021--Gaps have changed since this grant was awarded in FY2015; see Human Research Roadmap https://] including: Gap 1 (Need to identify and quantify the key threats to and promoters of mission relevant behavioral health and performance during exploration class missions) and Gap 3 (Need to identify and validate measures to monitor behavioral health and performance and determine acceptable thresholds for these measures during exploration missions). Our primary goal is to identify the psychological and behavioral health symptoms with the greatest likelihood of occurrence during extended human space flight/habitation to space and to estimate associated levels of threat imposed to mission-based performance. As a final deliverable, a checklist of symptoms will be developed for implementation during an ISS mission (>6 months) in order to determine its feasibility, reliability, and facilitation of evidence-based decision making with regard to crew health, safety, and mission success. As a first step, we will conduct extensive scientific literature reviews and interviews with subject matter experts in order to synthesize existing knowledge of the psychological and behavioral symptoms experienced in space and other extreme environments (Aim 1). Our review will directly inform the development of a comprehensive checklist of symptoms to be monitored among 8 separate cohorts (i.e., 4 Antarctic and 4 HERA cohorts) as part of a longitudinal investigation (Aim 2). Symptoms will be examined based on their point/period prevalence, severity, and duration. The checklist will also be administered (repeatedly) in conjunction with the Space Medicine Exploration Medical Condition List (SMEMCL) in order to examine concurrent and sequential overlap between psychological and physical health symptoms as means of clarifying potential etiologies. Finally, our study will extend previous research by exploring relationships among psychological health, sleep loss/dysregulation, biomarkers of stress, and performance-based outcomes (Aim 3). A comprehensive battery of cognitive and performance measures (including a perception-action coupling task) will be administered repeatedly as part of our longitudinal study. These outcomes will inform a final list of psychological/ behavioral symptoms to be examined during an extended ISS mission.

Task Progress:	During the past year, we published several papers from our Antarctic study, including primary mental health outcomes. Among the three Mental Health Checklist (MHCL) subscales, both positive adaptation and poor self-regulation scores changed significantly across the 9-month mission. For positive adaptation, a continuous linear decrease was observed from baseline to the end of the mission. Thus, even as crew members were preparing to return home, positive adaptation scores did not evidence a 'bounce back' effect. Deterioration in the range of items that make up this MHCL subscale (e.g., proud, inspired, enthused/spirited, interested/fascinated, perfectionistic) might be viewed as progressive feelings of detachment or indifference over time. For MHCL poor-regulation scores, the greatest increases were observed during the first half of the mission. Scores decreased somewhat in later months, though not back to baseline levels. Interestingly, higher poor self-regulation scores were observed early in the mission at McMurdo compared to the South Pole station. Considering the harsher, more extreme conditions at the South Pole, and the absence of station-based differences in demographic, military, or prior polar experience, poorer self-regulation scores at McMurdo may reflect the absence of pre-deployment psychological screening at this station. By comparison, monthly MHCL anxious apprehension scores did not change significantly over time. Thus, whereas declines in both positive emotion and self-regulatory abilities may be more universal experiences in this ICE setting, changes in anxiety might be shaped more so by individual (e.g., physical symptoms), situational, and/or interactional effects than by broader environmental stressors. We did observe persistently higher MHCL anxious apprehension scores at McMurdo station where weather elements are less extreme but the population is much larger. A potentially important direction for future work is to examine whether and how reports of anxious apprehension are affected by acute interpersonal stressors. We also examined changes in the severity of physical symptoms across mission, finding significant increases from baseline to the end of study at both stations. Elevation in physical symptoms during mission also predicted end of study poor self-regulation and anxious apprehension. Monthly levels of cortisol and cortisol:DHEA ratios were higher in McMurdo compared to South Pole participants indicating greater experiences of stress at the coastal station. Again, the larger McMurdo population likely influences levels of psychosocial stress; laboratory-based studies incorporating social stress paradigms have routinely been shown to evoke a cortisol response. In conjunction with higher MHCL anxious apprehension scores, findings are suggestive of higher levels of arousal/activation at McMurdo. Assessment of within-mission use of regulatory strategies indicated that savoring, reappraisal, and self-focused rumination were more often used in response to positive emotions than dampening and suppression. This pattern of responding is generally considered adaptive, as the former strategies are associated with effectively increasing and maintaining positive affect. Decreased use of suppression over time was also associated with greater end of study positive adaptation scores, aligning with community-based findings showing that, in addition to being cognitively taxing, suppression results in worsened emotional outcomes. Interestingly, savoring was significantly reduced late in the mission and at the end of the study, when positive adaptation scores were lowest. We also examined whether several trait-based constructs assessed at baseline predicted MHCL scores at the end of the study. Controlling for baseline positive adaptation scores, higher baseline Difficulties with Emotion Regulation Scale (DERS) scores predicted less positive adaptation at the end of the study. Overall, our findings add to a limited body of literature regarding mental health changes in the Antarctic. In all subjects, positive adaptation scores progressively decreased across the study, suggestive of increasing feelings of detachment and/or indifference over time. Conversely, poor self-regulation scores increased across the study and were predicted by co-occurring increases in the severity of physical symptoms. One particularly novel aspect of our study was the assessment of within-mission use of strategies for regulating positive emotion. These data suggest that greater expression of positive emotions and savoring positive experiences may provide a buffer against typical decreases in positive adaptation that occur over long periods in ICE environments. Interventions and/or countermeasures directed at sustaining and enhancing positive emotions across extended missions may therefore be useful tool for reducing psychological risk.
Bibliography Type:	Description: (Last Updated: 12/23/2022)
Abstracts for Journals and Proceedings	Cifre AB, Kim J, So CJ, Gonzalez R, Abeln V, Alfano CA. "Global Sleep Quality as a Predictor of Mental Health During HERA Missions." Poster presented at the 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Kim J, Cifre AB, So CJ, Gonzalez R, Bower J, Connaboy C, Simpson R, Alfano CA. "Markers of Distress Among Medical and Behavioral Health Evacuees from Antarctica." Poster presented at the 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Alfano C, Mi Q, Ahamed NU, Bower JL, Connaboy C, Simpson RJ. "Sleep Regularity and Mental Health during Extended Antarctic." Paper presented at the 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Articles in Peer-reviewed Journals	Alfano CA, Bower JL, Connaboy C, Agha NH, Baker FL, Smith KA, So CJ, Simpson RJ. "Mental health, physical symptoms and biomarkers of stress during prolonged exposure to Antarctica's extreme environment." Acta Astronautica. 2021 Apr;181:405-13. https://doi.org/10.1016/j.actaastro.2021.01.051 , Apr-2021
Articles in Peer-reviewed Journals	Connaboy C, Sinnott AM, LaGoy A, Krajewski KT, Johnson CA, Pepping G-J, Simpson RJ, Bower JL, Alfano CA. "Cognitive performance during prolonged periods in isolated, confined, and extreme environments." Acta Astronautica. 2020 Dec;177:545-51. https://doi.org/10.1016/j.actaastro.2020.08.018 , Dec-2020
Articles in Peer-reviewed Journals	LaGoy A, Sinnott AM, Ambarian M, Pepping G-J, Simpson RJ, Agha NH, Bower JL, Alfano CA, Connaboy C. "Differences in affordance-based behaviors within an isolated and confined environment are related to sleep, emotional health and physiological parameters." Acta Astronautica. 2020 Nov;176:238-46. https://doi.org/10.1016/j.actaastro.2020.06.034 , Nov-2020